Fritz Family Fellows
Public Health Needs & Privacy Issues

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Bansal Lab

MASSIVE DATA INSTITUTE

GEORGETOWN UNIVERSITY College of Arts & Sciences | Department of Computer Science

Cross disciplinary research between epidemiology and privacy

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Support

Fritz Family Fellowship

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Mobile device location data to track human mobility

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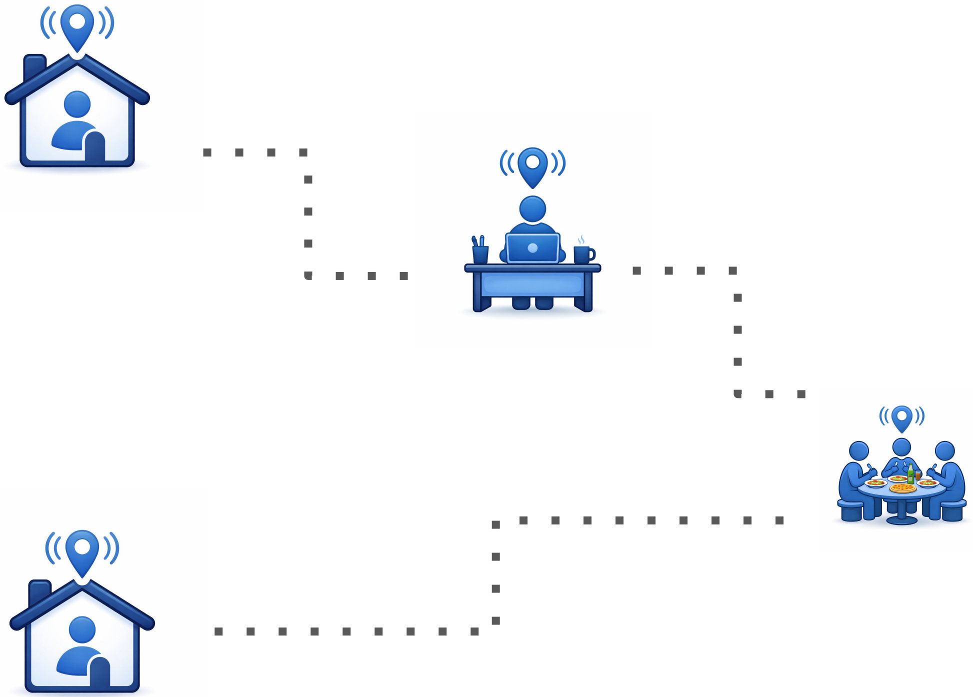

Privacy concerns on access to human mobility data

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SAFEGRAPH

veraset

unacast.

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FBI is buying data that can be used to track people, Patel says

This is the first confirmation that the FBI has resumed actively buying people's data for investigations.

03/18/26, *POLITICO*

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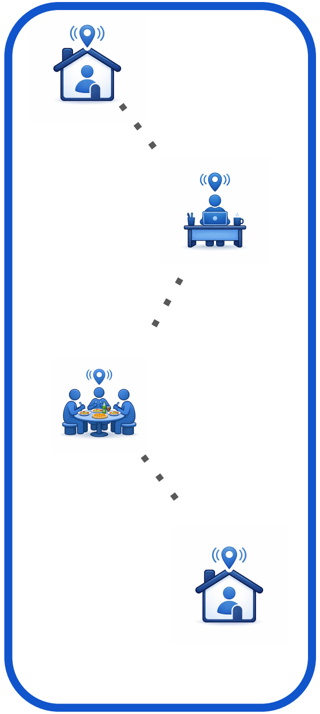
Location data broker SafeGraph stops selling information on visits to abortion providers

05/13/22, *CNBC*

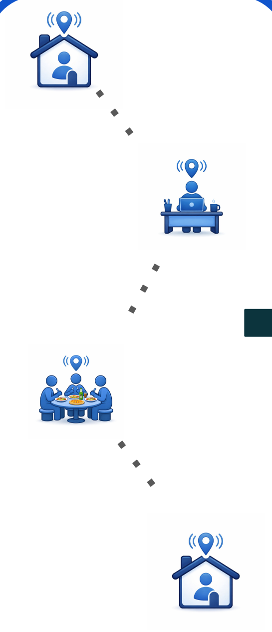
Aggregated mobility data used in epidemiological research

Individual data

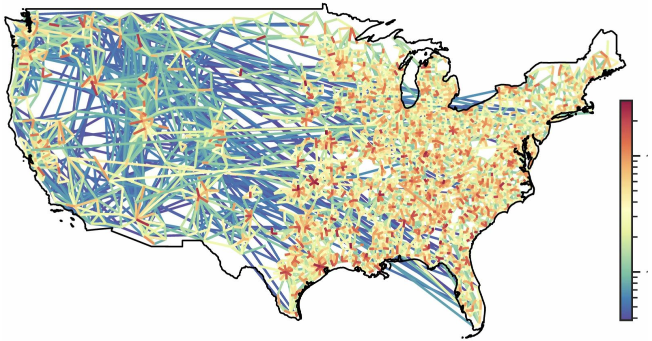


Aggregated mobility data also used in epidemiological research

Individual data



Aggregated data



Individual data aggregated at county or CBG level

Pullano et al.,
JMIR Public Health and Surveillance, 2025

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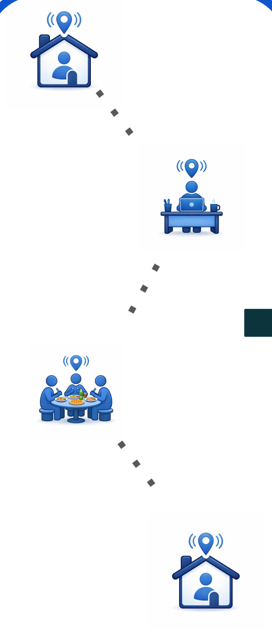

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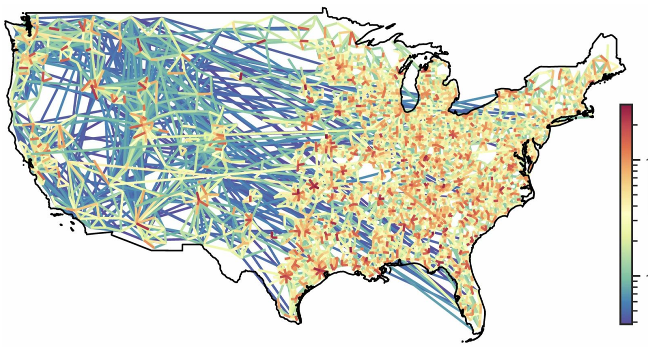

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Pullano et al.,
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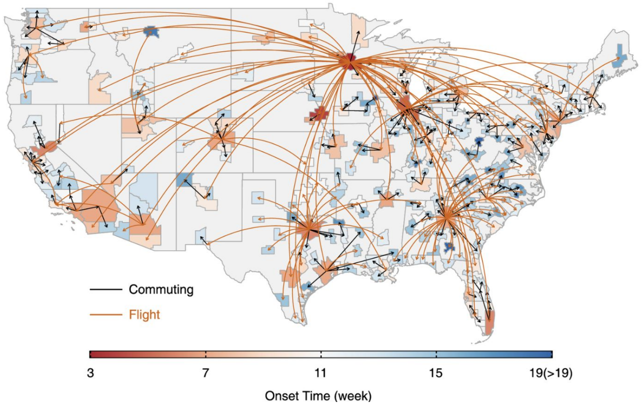
Disease spread modeling

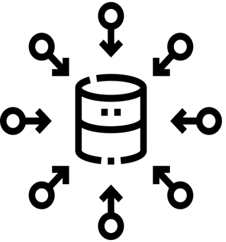


How infectious disease spread and peak at regional or national level

Zhang et al.,
Population Biology, 2026

Do existing privacy protections in epidemiology provide sufficient privacy?

Pseudonymization:
Remove clearly identifying information (e.g., name, address, SSN)



Aggregation:
Provide aggregated statistics (e.g., county, CBG level)

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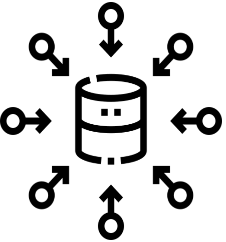

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Are aggregation and pseudonymization sufficient?

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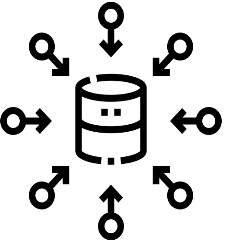
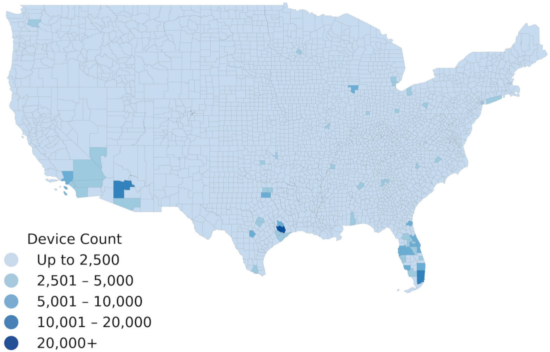


Aggregation:
Provide aggregated statistics (e.g., county, CBG level)

Are aggregation and pseudonymization sufficient?

Can other privacy-preserving methods reduce risk and maintain epidemiological accuracy?

Data overview and access



Device Count
(National)
3,229 counties
365 days (2019)
Approx 11 million daily devices

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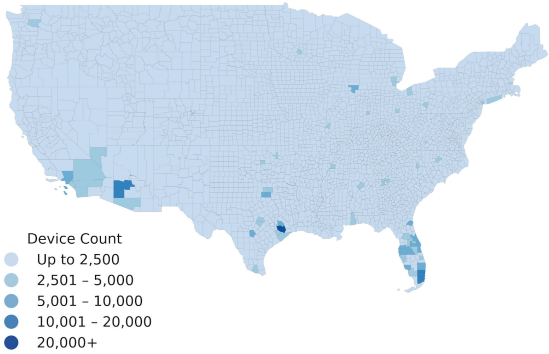
Data overview and access



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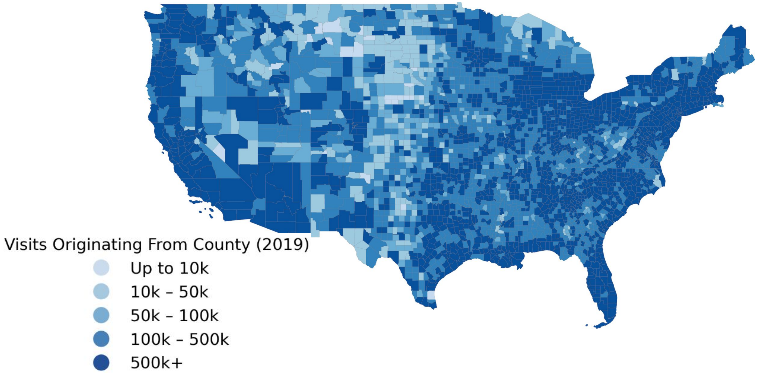

County to County Visits
(National)
3,229 counties
365 days (2019)
6,345 million visits

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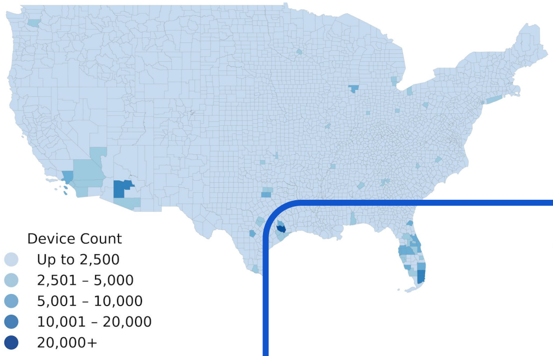
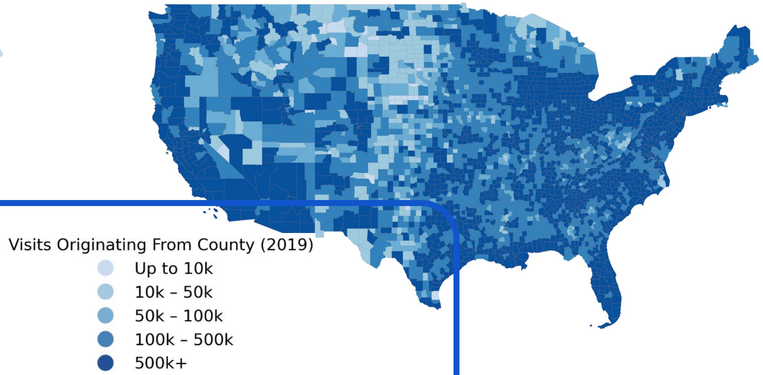
Data overview and access*





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Privacy by platform:
- No individual traces
- Aggregated at county level
- Masked counts for low # of visits

** Also generated synthetic individual traces using a gravity model*

Membership Inference Attack

Designed a game to test privacy risks

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Define Release:

Average probability of visiting a county

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Attack Scenario:

Access to individual traces and and release

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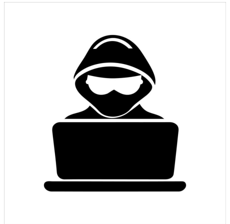

Access to individual traces and and release

Guess:

Guess whether trace used in release or not

Membership Inference Attack

Designed a game to test privacy risks

Define Release:

Average probability of visiting a county

Attack Scenario:

Access to individual traces and and release

Guess:

Guess whether trace used in release or not

Correctly guess more than 50% of the time: PRIVACY RISK

Implementing the membership inference attack

Compared results from two counties with different mobility patterns

County Case Studies

County 1:

High device, high within county movement

County 2:

Low device, low within county movement

March 2019

Met criterion on # devices

Implementing the membership inference attack

County Case Studies

<u>County 1:</u>
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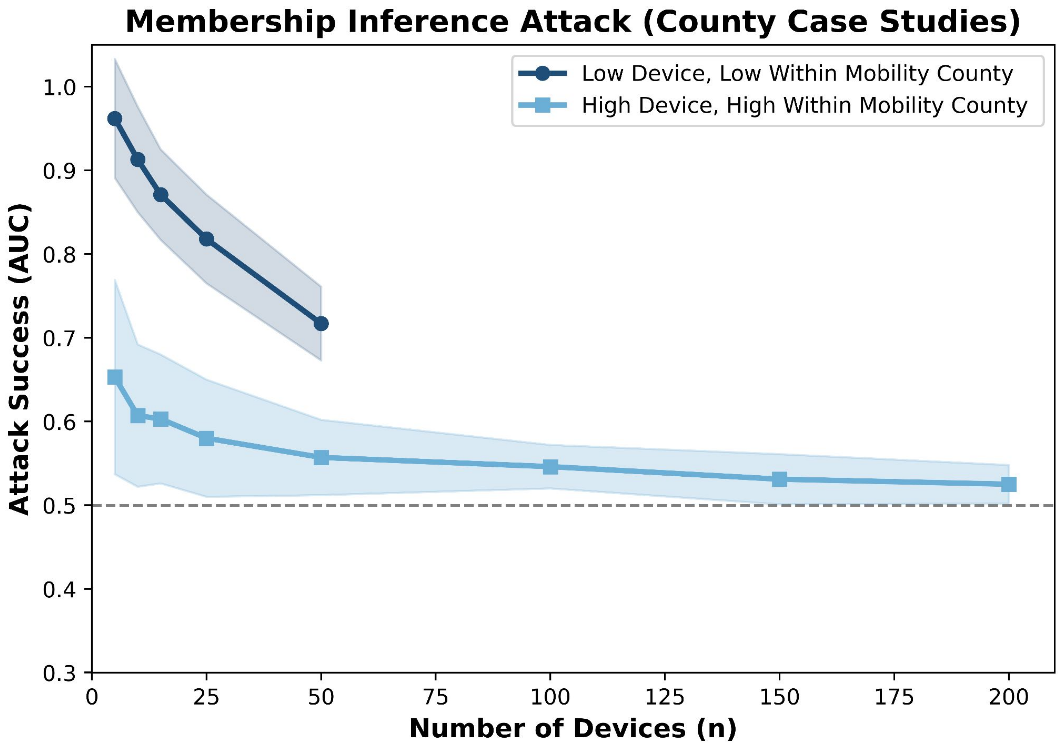


Membership inference risks in aggregated, pseudonymized mobility data

Dealing with privacy risks in mobility data

Membership inference in aggregated, pseudonymized mobility data

Differentially private disease spread modeling

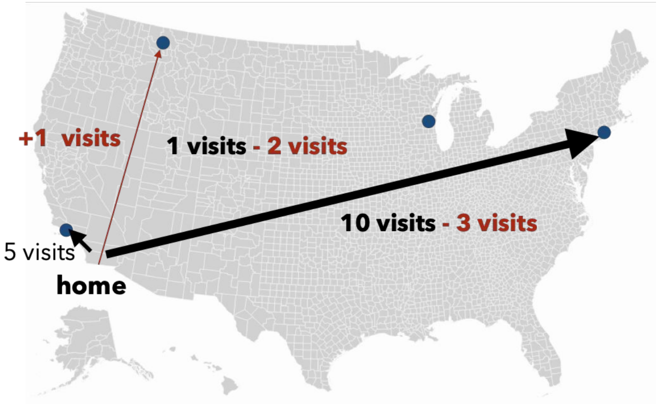
Using differential privacy to mask individual contributions

Differential privacy provides mathematical guarantees that any individual's inclusion or exclusion has 'small' effect on statistical outputs

aggregate data + $Lap(0, \frac{k}{\epsilon})$

max visits per individual = 10 (from data)

privacy parameter



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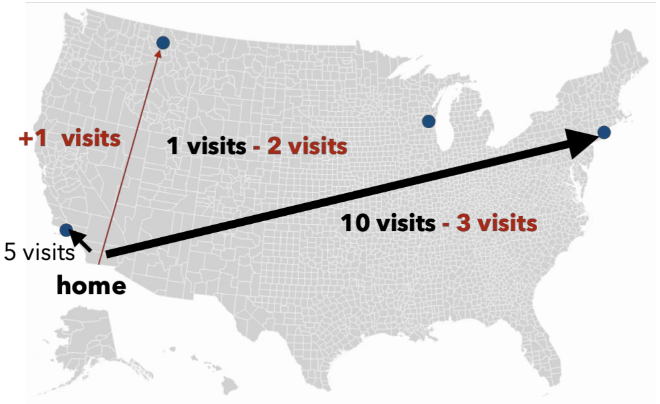

<u>Minimization</u>:

Removed high mobility devices (visit count > 10)

Effect of DP noise on mobility structure

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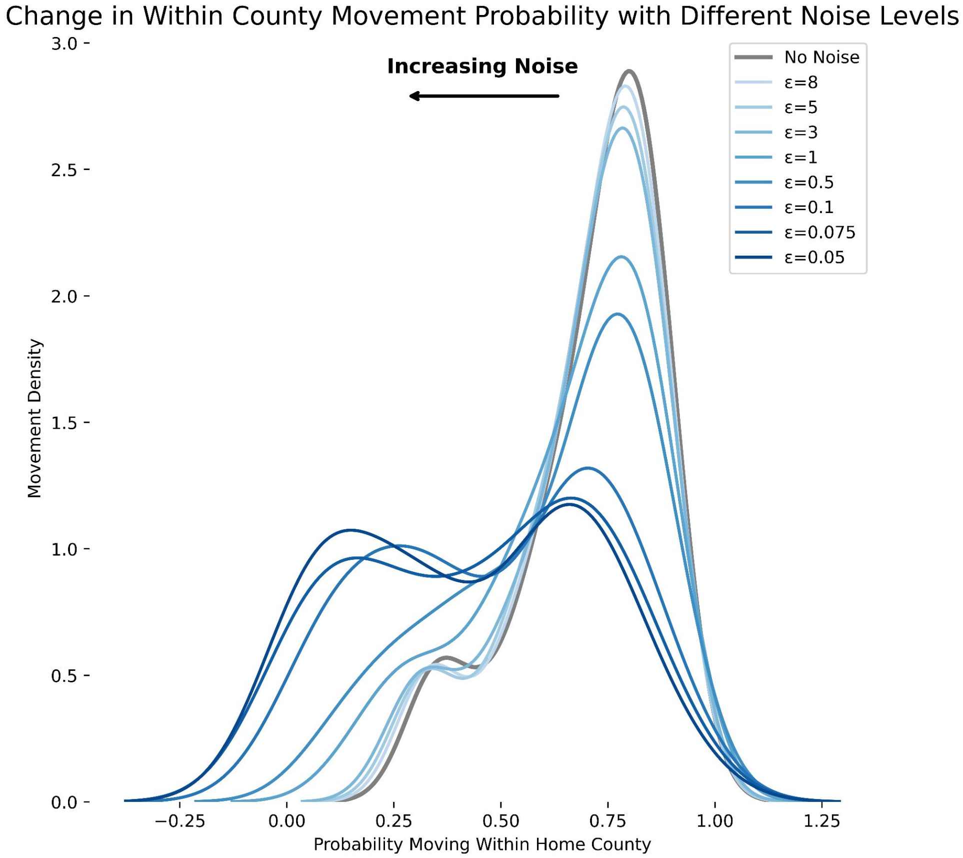
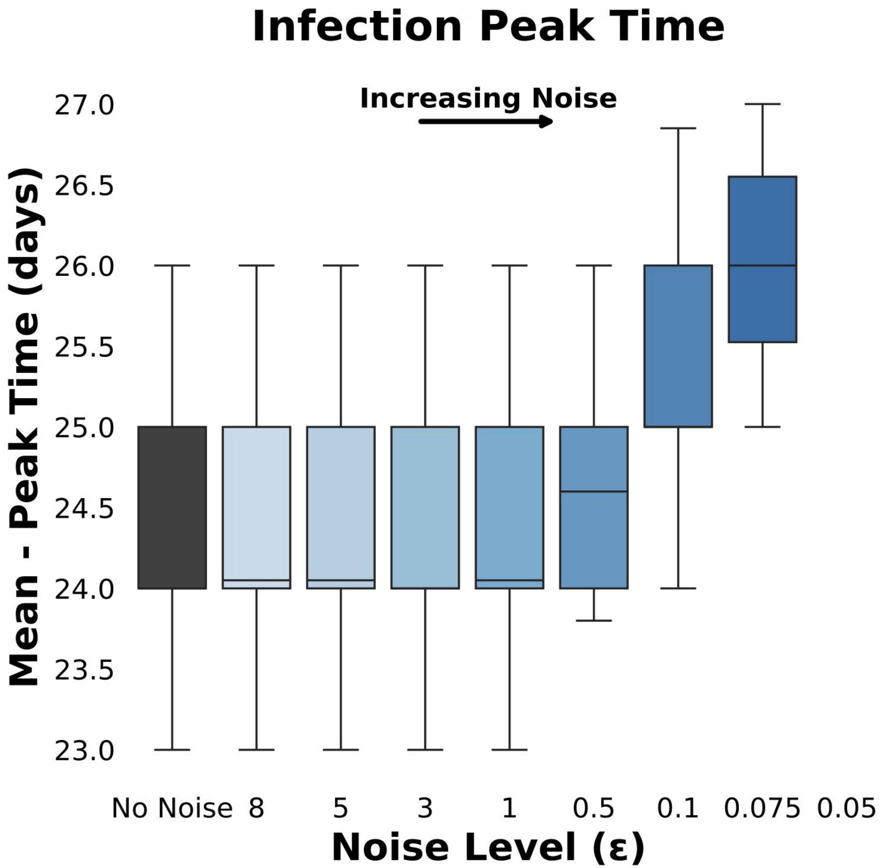


Effect on disease spread modeling accuracy

When cases peak in a community



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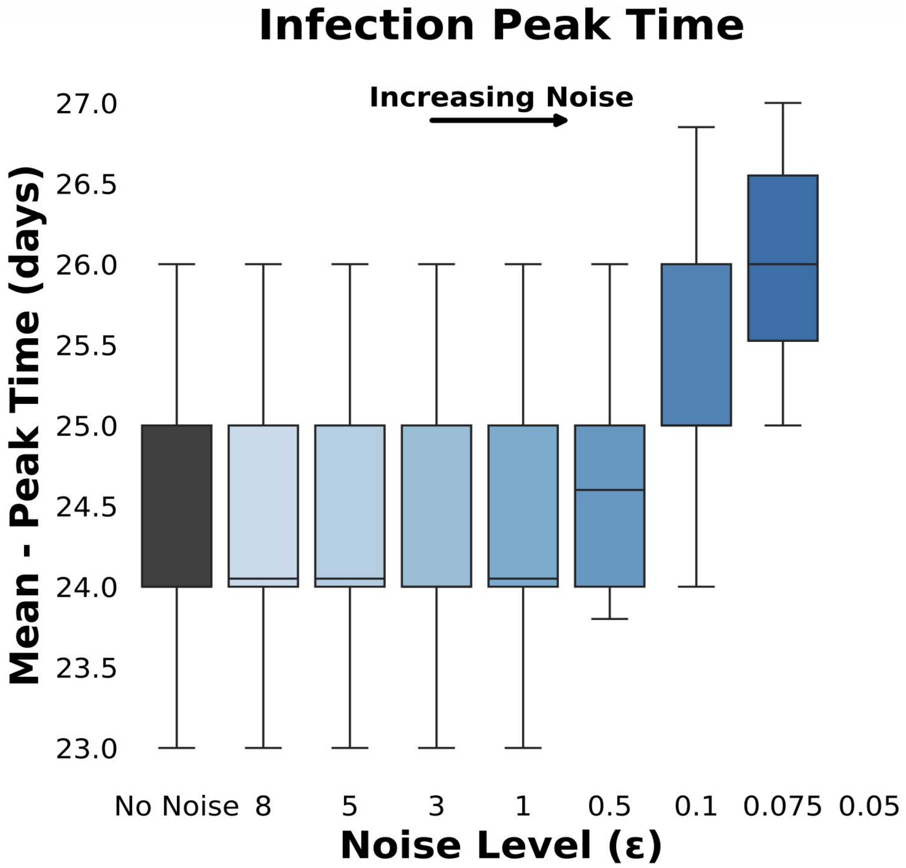

Balance spread accuracy and privacy **for high noise levels**

Conclusion

Privacy risks for pseudonymized and aggregated mobility data

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Individuals in low device and atypical mobility counties at risk

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Balance epidemiological accuracy and privacy at high noise levels

Conclusion

Privacy risks for pseudonymized and aggregated mobility data

Individuals in low device and atypical mobility counties at risk

Balance epidemiological accuracy and privacy at high noise levels

Finalize research findings and publish manuscript!

References

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Thank you!